

QF620 Stochastic Modelling in Finance

Assignment 1/4

Due Date: 25-Sep-2024

1. Let $X \sim N(\mu, \sigma^2)$ be a random variable following normal distribution. Use the Moment Generating Function to evaluate the following expectation:

(a) $\mathbb{E}[e^X]$.

(b) $\mathbb{E}[e^{2X}]$.

①
$$\begin{aligned} \mathbb{E}(e^x) &= \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{\infty} e^x \cdot e^{-\frac{(x-\mu)^2}{2\sigma^2}} \cdot dx \\ &= \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{\infty} e^{x - \frac{x^2 - 2\mu x + \mu^2}{2\sigma^2}} \cdot dx \\ &= \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{\infty} e^{-\frac{-2\sigma^2 x + x^2 - 2\mu x + \mu^2}{2\sigma^2}} \cdot dx \\ &= \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{\infty} e^{-\frac{x^2 - 2(\mu + \sigma^2)x + \mu^2}{2\sigma^2}} \cdot dx \\ &= \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{\infty} e^{-\frac{x^2 - 2(\mu + \sigma^2)x + (\mu^2 + 2\mu\sigma^2 + \sigma^4) - 2\mu\sigma^2 - \sigma^4}{2\sigma^2}} \cdot dx \\ &= \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{\infty} e^{-\frac{x^2 - 2(\mu + \sigma^2)x + (\mu + \sigma^2)^2}{2\sigma^2} + \mu + \frac{\sigma^2}{2}} \cdot dx \\ &= \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{\infty} e^{-\frac{[x - (\mu + \sigma^2)]^2}{2\sigma^2} + \mu + \frac{\sigma^2}{2}} \cdot dx \\ &= e^{\mu + \frac{\sigma^2}{2}} \cdot \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{\infty} e^{-\frac{(x - (\mu + \sigma^2))^2}{2\sigma^2}} \cdot dx \\ &= e^{\mu + \frac{\sigma^2}{2}} \end{aligned}$$

②
$$\begin{aligned} \mathbb{E}(e^{2x}) &= \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{\infty} e^{2x} \cdot e^{-\frac{(x-\mu)^2}{2\sigma^2}} \cdot dx \\ &= \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{\infty} e^{2x - \frac{x^2 - 2\mu x + \mu^2}{2\sigma^2}} \cdot dx \\ &= \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{\infty} e^{-\frac{-2\sigma^2 \cdot 2x + x^2 - 2\mu x + \mu^2}{2\sigma^2}} \cdot dx \\ &= \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{\infty} e^{-\frac{x^2 - 2(\mu + 2\sigma^2)x + \mu^2}{2\sigma^2}} \cdot dx \\ &= \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{\infty} e^{-\frac{x^2 - 2(\mu + 2\sigma^2)x + (\mu^2 + 4\mu\sigma^2 + 4\sigma^4) - 4\mu\sigma^2 - 4\sigma^4}{2\sigma^2}} \cdot dx \\ &= \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{\infty} e^{-\frac{x^2 - 2(\mu + 2\sigma^2)x + (\mu + 2\sigma^2)^2}{2\sigma^2} + 2\mu + \frac{4\sigma^2}{2}} \cdot dx \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{\infty} e^{-\frac{[\kappa - (\mu + 2\sigma^2)]^2}{2\sigma^2} + 2\mu\kappa + \frac{4\sigma^2}{2}} d\kappa \\
&= e^{2\mu + \frac{4\sigma^2}{2}} \cdot \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{\infty} e^{-\frac{(\kappa - (\mu + 2\sigma^2))^2}{2\sigma^2}} d\kappa \\
&= e^{2\mu + \frac{4\sigma^2}{2}} \\
&= e^{2\mu + 2\sigma^2}
\end{aligned}$$

2. Let $W_t \sim N(0, t)$ be a random variable following normal distribution. Use the Moment Generating Function to evaluate the following expectation:

(a) $\mathbb{E}[e^{W_t}]$.

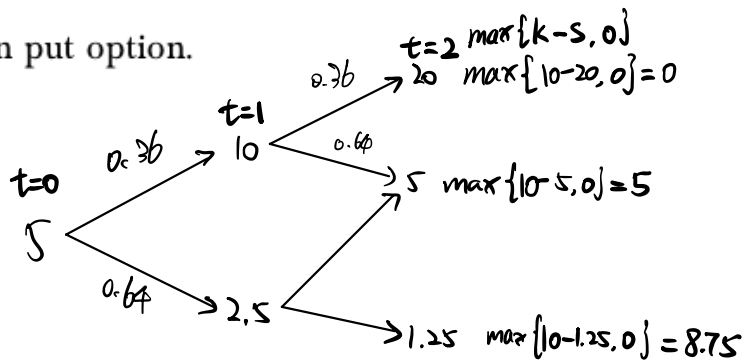
(b) $\mathbb{E}[e^{\sigma W_t}]$, where σ is a constant.

① $E(e^{W_t}) = M_{W_t}(1) = e^{\mu + \frac{1}{2}t} = e^{\frac{1}{2}t}$

② $E(e^{\sigma W_t}) = M_{W_t}(\sigma) = e^{\mu\sigma + \frac{1}{2}t\sigma^2} = e^{\frac{1}{2}t\sigma^2}$

3. Consider a Cox-Ross-Rubinstein binomial tree with $S_0 = \$5$, $u = 2$, $r = 4\%$. We would like to value a vanilla option with a strike price of $K = \$10$, maturing on the second time step, i.e. $t = 2$. Calculate the option price if the exercise style and payoff are:

(a) European put option.



$$d = \frac{1}{u} = \frac{1}{2} = 0.5$$

$$p^* = \frac{(1+r)-d}{u-d} = \frac{(1+0.04)-0.5}{2-0.5} = 0.36$$

$$\text{Put} = \frac{1}{(1+r)^2} \cdot E^*(P_2)$$

$$= \frac{1}{(1+0.04)^2} [0.36 \times 0.36 \times 0 + 0.36 \times 0.64 \times 5 + 0.64 \times 0.64 \times 8.75]$$

$$= \frac{1}{1.04^2} (2.304 + 3.584)$$

$$= 5.44$$

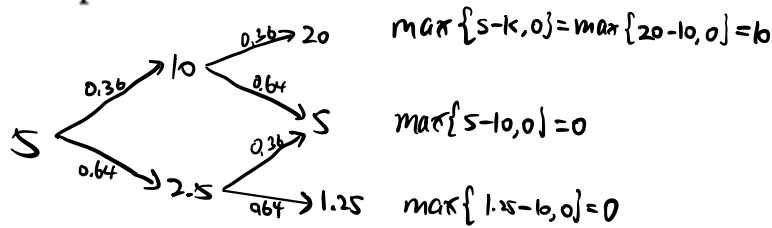
(b) American put option.

$$\max \left\{ 10 - 10, \frac{1}{1.04} (0.64 \times 5 + 0.36 \times 0) \right\} = 3.08$$

$$\max \left\{ 10 - 2.5, \frac{1}{1.04} (0.36 \times 5 + 0.64 \times \frac{5}{4}) \right\} = 7.5$$

$$\max \left\{ 10 - 5, \frac{1}{1.04} (0.36 \times 3.08 + 0.64 \times 7.5) \right\} = 5.08$$

(c) European call option.



when $t=1, S_1=10$

$$V_u^E = \frac{1}{1+0.04} (0.36 \times 10 + 0.64 \times 0) = 3.46$$

when $t=1, S_1=2.5$

$$V_d^E = \frac{1}{1+0.04} (0.36 \times 0 + 0.64 \times 0) = 0$$

when $t=0, S_0=0$

$$V_0^E = \frac{1}{1+0.04} (3.46 \times 0.36 + 0 \times 0.64) = 1.2$$

(d) American call option.

$$\max \left\{ 10 - 10, \frac{1}{1.04} (0.36 \times 10 + 0.64 \times 0) \right\} = 3.46$$

$$\max \left\{ 2.5 - 10, \frac{1}{1.04} (0.36 \times 0 + 0.64 \times 0) \right\} = 0$$

$$\max \left\{ 5 - 10, \frac{1}{1.04} (0.36 \times 3.46 + 0 \times 0.64) \right\} = 1.2$$

4. Let W_t denote a standard Brownian motion. Calculate the following probabilities:

(a) $\mathbb{P}(W_2 < 0 | W_1 > 0)$

(b) $\mathbb{P}(W_1 \times W_2 < 0)$

(c) $\mathbb{P}(W_1 < 0 \cap W_2 < 0)$

$$W_1 \sim N(0, 1)$$

$$W_2 \sim N(0, 2)$$

$$\because W_2 = W_1 + (W_2 - W_1) \quad \therefore W_2 - W_1 \sim N(0, 1)$$

(a) $P(W_2 < 0 | W_1 > 0)$

$$= P(\text{step down} \cap \text{2nd step bigger than 1st step} | W_1 > 0)$$

$$= P(\{W_2 < W_1\} \cap \{W_2 - W_1 > -W_1\} | W_1 > 0)$$

$$= P(\{W_2 < W_1\} | W_1 > 0) \cdot P(\{W_2 - W_1 > -W_1\} | W_1 > 0)$$

$$= 0.5 \times 0.5$$

$$= 0.25$$

$$\textcircled{b} P(W_1 \times W_2 < 0)$$

the sign of W_1 and W_2 are opposite.

$$= P(W_1 > 0 \cap W_2 < 0) + P(W_1 < 0 \cap W_2 > 0)$$

$$= P(W_1 > 0) \cdot P(W_2 < 0 | W_1 > 0) + P(W_1 < 0) \cdot P(W_2 > 0 | W_1 < 0)$$

$$= 0.5 \times 0.25 + 0.5 \times 0.25$$

$$= 0.125 + 0.125$$

$$= 0.25$$

$$\textcircled{c} P(W_1 < 0 \cap W_2 < 0)$$

$$= P(W_1 < 0 | W_2 < 0) \cdot P(W_2 < 0)$$

$$= 0.5 \times 0.75$$

$$= 0.375$$

5. Let W_t denote a standard Brownian motion. Let $s < t$, determine the variance of

$$(a) V[(W_t - W_s)].$$

$$(b) V[(W_t - W_s)^2].$$

$$\textcircled{a} V[(W_t - W_s)]$$

$$= E[(W_t - W_s)^2] - E[W_t - W_s]^2$$

$$'\because W_t - W_s \sim N(0, t-s)$$

$$= t-s - 0^2$$

$$= t-s$$

$$\textcircled{b} \quad V[(W_t - W_s)^2]$$

$$= E[(W_t - W_s)^4] - (E[(W_t - W_s)^2])^2$$

$$\because E(W_t - W_s) = 0, \quad W_t - W_s \sim N(0, t-s)$$

$$E[(W_t - W_s)^2] = t-s$$

$$X \sim N(0, \sigma^2)$$

$$= 3(t-s)^2 - (t-s)^2$$

$$= 2(t-s)^2$$

6. Let W_t denote a standard Brownian motion. Evaluate the following expectation

$$E[|W_{t+\Delta t} - W_t|],$$

where $|\cdot|$ denote absolute value.

According to Brownian motion;

we know that $W_{t+\Delta t} - W_t \sim N(0, \Delta t)$,

then, standardize it, we have:

$$Z = \frac{W_{t+\Delta t} - W_t - 0}{\sqrt{\Delta t}}$$

$$W_{t+\Delta t} - W_t = Z \cdot \sqrt{\Delta t}$$

$$\therefore Z \sim N(0, 1)$$

$$E[|W_{t+\Delta t} - W_t|] = E[|\sqrt{\Delta t} \cdot Z|]$$

$$= \sqrt{\Delta t} \cdot E(|Z|)$$

$$E(|Z|) = \int_{-\infty}^{\infty} |z| \cdot f_z(z) \cdot dz$$

$$= 2 \int_0^{\infty} |z| \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} dz$$

$$\text{let } x = \frac{z^2}{2}, \quad dx = z \cdot dz$$

$$= 2 \int_0^{\infty} \frac{1}{\sqrt{2\pi}} e^{-x} \cdot dx$$

$$= 2 \cdot \frac{1}{\sqrt{\pi}} \int_0^{\infty} e^{-x^2} dx$$

$$= 2 \cdot \frac{1}{\sqrt{\pi}}$$

$$= \frac{2}{\sqrt{\pi}}$$

$$= \sqrt{\frac{2}{\pi}}$$

plug $E(|Z|)$ into $E[|W_{t+\Delta t} - W_t|]$

So we get $\sqrt{\Delta t} \cdot E(|Z|)$

$$= \sqrt{\Delta t} \cdot \sqrt{\frac{2}{\pi}}$$

$$= \sqrt{\frac{2\Delta t}{\pi}}$$